

codestorywithmiK

CSwithMIK

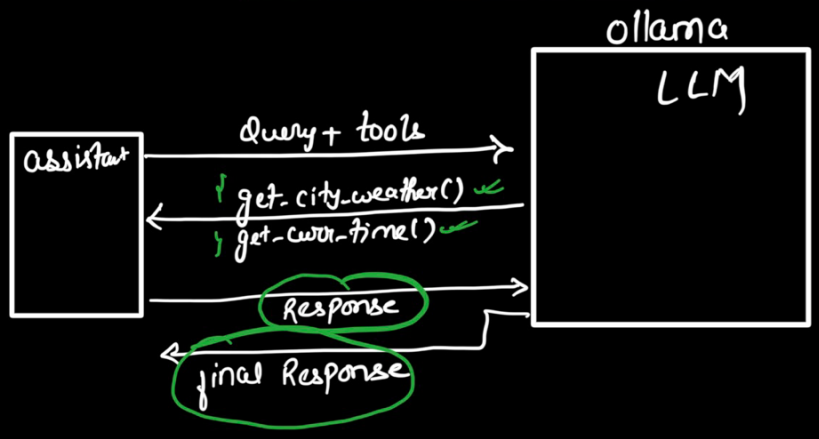
codestorywithMIK

Chained Tool calling



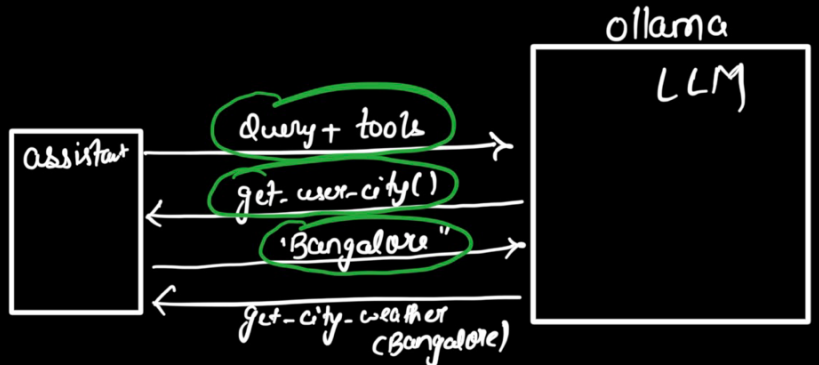
Case-1 : What is the weather in Bangalore ? and what is the current time ?

Parallel Tool Calls

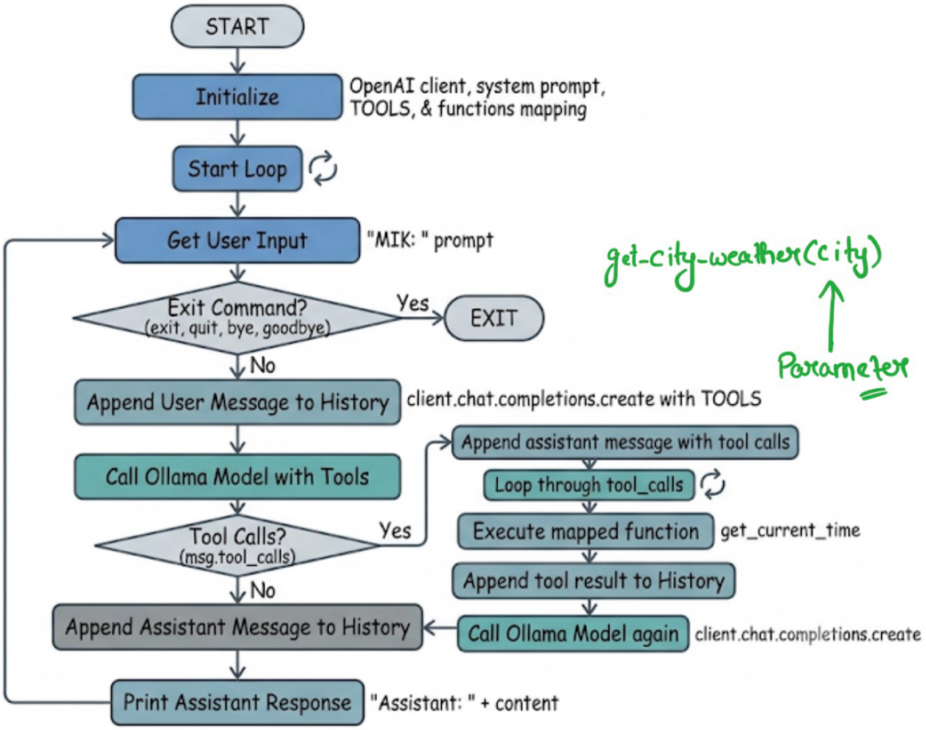


Case-2: What is the weather of the city where I live ? Chained (sequential) tool calls.

Agent.



Recap : video - 8

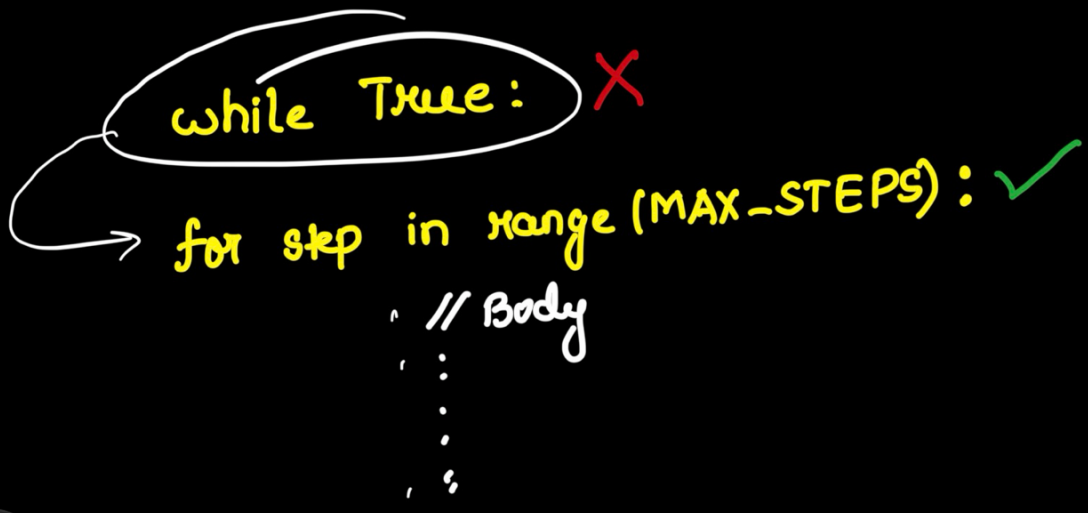


Lets do some quick experiments

- How to handle?
  Put a Limited for loop instead of while True:

② what if any of our tools crashes ?

(·) Put the tool call inside try catch.
(·) If error comes, send that error to model as the tool result.
Model can the explain the problem politely to the user.



```
messages.append({
    "role": "assistant",
    "content": msg.content
})
print("Assistant: " + msg.content)
```

Let's test this out

Two dangers in our code...

① `while True:` // chained ToolCalling

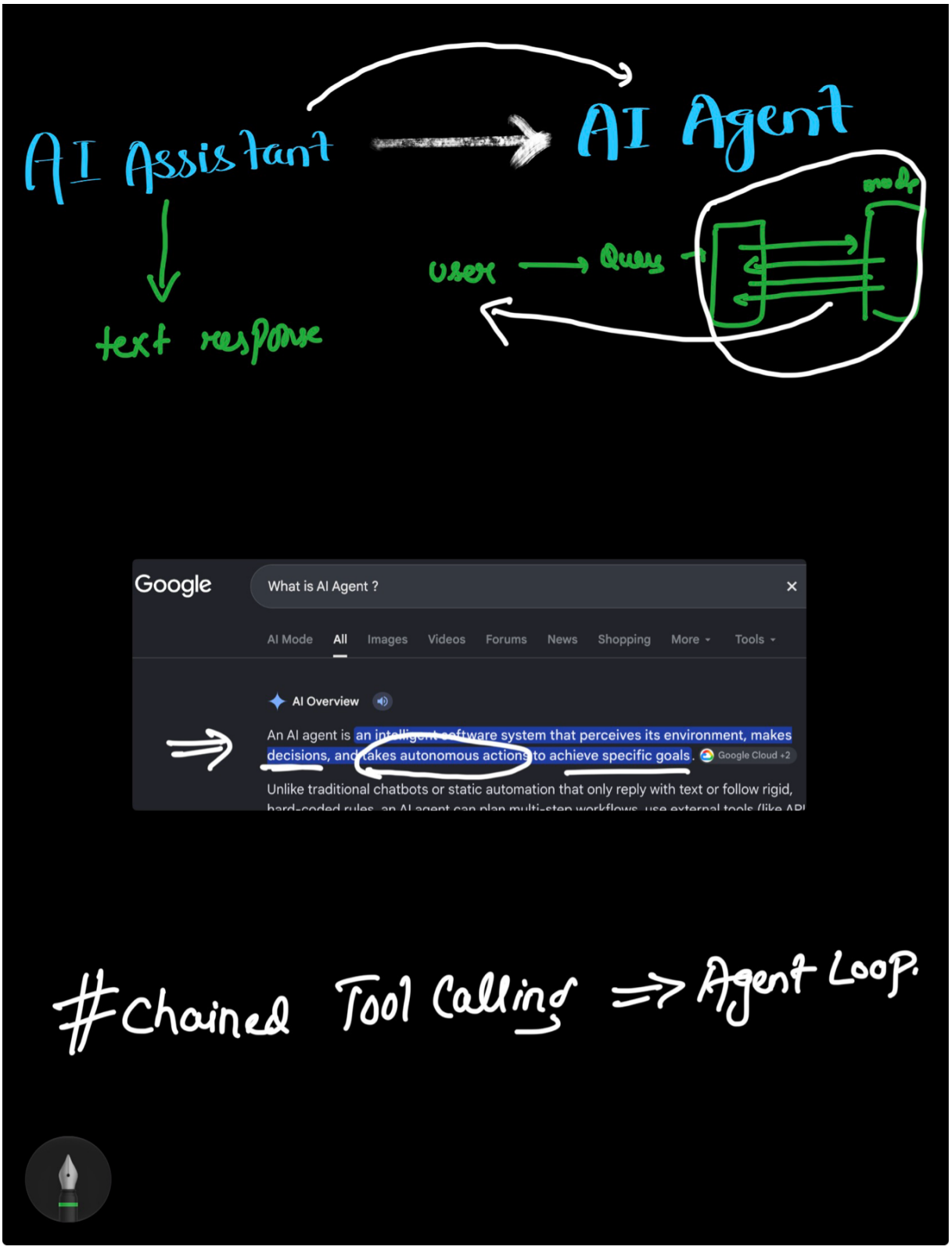
- What if model keeps asking for tool calls. It will run forever.



- How to handle?
  Put a Limited for loop instead of while True:

```
65          ) model=os.getenv("MODEL"),
66              messages=messages,
67          )
68
69          msg = response.choices[0].message
70
71      messages.append({
72          "role" : "assistant",
73          "content" : msg.content
74      })
75
76      print("Assistant: " + msg.content)
```

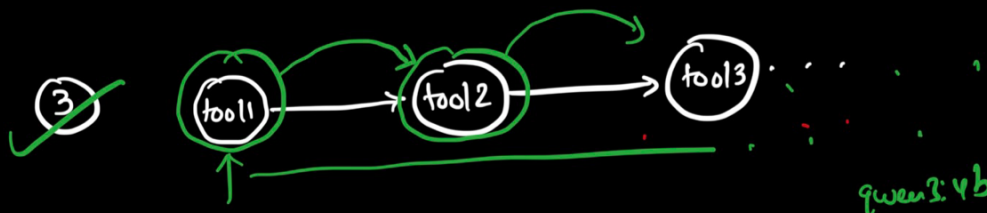



AI Assistant
AI Agent
text response
user
Query
Google
What is AI Agent ?
AI Mode
All
Images
Videos
Forums
News
Shopping
More
Tools
AI Overview
An AI agent is an intelligent software system that perceives its environment, makes decisions, and takes autonomous actions to achieve specific goals.
Google Cloud +2
Unlike traditional chatbots or static automation that only reply with text or follow rigid,
#Chained Tool Calling => Agent Loop

Limitations

1. Every Tool Calling (for loop) costs time & (token)

2. Chained tool calling can run indefinitely.
   MAX_RUN_STEPS

3. 

   qwen3:4b

4. Small models struggle with (multi-step planning), can call tools in (wrong order).

5. Since LLM is (non-deterministic), it becomes (difficult to debug).

6. (Prompt injection)
   ↳ Example:- tool → read (web-page)

```python
for tool_call in msg.tool_calls:
    func_name = tool_call.function.name
    func_args = json.loads(tool_call.function.arguments or "{}")

    fn = FUNCTION_NAME_TO_FUNCTION_MAP[func_name]
    result = fn(**func_args)

    messages.append({
        "role": "tool",
        "tool_call_id" : tool_call.id,
        "content" : str(result),
    })
```



```
for tool_call in msg.tool_calls:
    func_name = tool_call.function.name
    func_args = { }

    try:
    {   function_args = json.loads(. . . . . .)
        fn = FUNCTION_NAME_TO_FUNCTION_MAP[func_name]
        result = fn(**func_args)
    except Exception as e :
        result = f"Error while running {func_name}
                 : {e}"
```

